MATERNAL HEALTH ANALYTICS DASHBOARD

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1. What This Tool Does

This dashboard puts together two different perspectives of maternal health so the team doesn't have to switch between spreadsheets.

One side is the big picture: World Health Organization (WHO) country data dating back to 1985, showing how maternal mortality changed globally.

The other side is closer to the ground: real patient vitals that the dashboard can use to flag a risk level on the spot.

This dashboard is designed to answer two kinds of questions at once, where is this problem worst? And what does it look like for one patient in front of you?

2. Getting In

Item	Detail
Live link	https://maternal-health-dashboard-pmu2wupa6nfyvcrmtthxtn.streamlit.app/
Password	maternalhealth2026
Works on	Desktop, tablet, and mobile browsers
Getting around	The sidebar on the left switches between the six pages

3. Where the Data Comes From

Behind this dashboard, are two different datasets that are kept separate intentionally.

One tracks countries over years, the other tracks individual patients at a single point in time.

Dataset	Source	Size	What it covers
WHO Maternal Mortality Ratio	WHO MNCAH Data Portal	185+ countries, 1985-2023	Maternal deaths per 100,000 births, broken down by country, region, and income level

Dataset	Source	Size	What it covers
Maternal Health Risk Dataset	Kaggle / UCI ML Repository	1,014 patient records	Real patient vitals - blood pressure, blood sugar, temperature, heart rate - each labeled with a risk outcome

4. Variable Glossary

WHO Dataset

Variable	Description
Year	Calendar year of the estimate, 1985 through 2023
Country / ISO 3 code	Country name and the 3-letter code used to plot the map
WHO region	One of six WHO regions
World Bank income group	Low, lower middle, upper middle, and high income
Value Numeric	The Maternal Mortality Ratio (deaths per 100,000 live births)
Value low / high	The lower and upper bounds of WHO's confidence interval

Patient Dataset

Variable	Description
Age	Patient age in years
SystolicBP / DiastolicBP	Blood pressure readings (mmHg)
BS	Blood sugar level (mmol/L)
BodyTemp	Body temperature (Fahrenheit)
HeartRate	Heart rate (bpm)
RiskLevel	Outcome label - low, mid, or high risk

5. What's on Each Page

Overview

The first thing you'll see. The headline figures in the dataset:

- Global average MMR,
- Reduction Since 1985

- Highest MMR
- Lowest MMR
- Countries tracked worldwide

Below that, a quick look at the ten worst-off countries and a breakdown by income categories, then the full trend line.

This page is intended to serve as a small summary for people who have only has thirty seconds, this is what they should see. It's the easiest page to read out loud in a meeting, since every number is labeled and color-coded without needing extra explanation.

Global Trends

This is where the narrative takes place over time. First, we have the global average which is represented by one line, second, we have a tool to pick any countries and stack them against each other, and a third chart splitting the trend by income group making it clearer how unevenly progress has landed.

Try comparing a low-income country against a high-income one here, the difference is shocking and is consistent almost every year in the dataset. That consistency is part of what makes the case for sustained, and long-term investment.

Geographic Map

A world map shaded by maternal mortality, with a slider to navigate through the years.

Hover any country for the exact number, and there's a table underneath if you want the raw figures for a given year.

Dragging the slider from 1985 to today is a quick way to show progress visually without saying a word, the map gets noticeably lighter across most areas, except where it remains stubbornly dark.

Regional Analysis

Same concept, sliced by WHO region and income levels instead of country.

The box plot is worth a look; it shows how wide the spread is within low-income countries compared to how tight it is in high-income countries.

The box's width matters as much as its position, a wide box means some countries in that group are doing far better or worse than others, which is a signal that averages alone can hide.

Patient Explorer

This is the clinical side. Filter by age or risk level in the sidebar and the three charts update, age spread by risk, a blood pressure versus blood sugar scatter, and the overall risk breakdown.

There's a searchable table at the bottom with the actual records.

This page is intended for a more clinical, hands-on audience. Someone who wants to poke at the actual numbers rather than just see a summary.

The scatter plot in particular is useful for spotting whether high-risk patients cluster in a recognizable range.

Risk Predictor

By typing in the patient's age, blood pressure, blood sugar, temperature, and heart rate, the model gives you the risk level (low, mid, or high) along with how accurate it is.

This Random Forest model with 82% accuracy on data it hasn't seen before. Blood sugar and systolic blood pressure turned out to matter most.

This is the page makes prediction rather than just showing data. It's a decision-support tool, not a diagnosis, the confidence chart is there so the result is never treated as a flat yes or no.

6. A Few Ways to Use It

- **Briefing leadership:** start on Overview, then jump to the map to show where things are worst.
- **Comparing countries or regions:** Global Trends or Regional Analysis will do that.
- **Checking a patient:** Risk Predictor takes their vitals and gives a read in seconds.
- **Digging into the patient data:** Patient Explorer, with the age and risk filters.

7. A Couple of Warnings

- WHO numbers are estimates with confidence intervals, not exact counts especially in countries where data collection is weaker.
- The patient dataset is 1,014 records. It's a useful sample, but the risk predictor is meant to support a clinician's judgment, not replace it.
- The data reflects the most recent download at the time this was built it isn't a live feed.